

Question ID: 312ba47c

| Assessment | Test | Domain                            | Skill                                                | Difficulty                                        |
|------------|------|-----------------------------------|------------------------------------------------------|---------------------------------------------------|
| SAT        | Math | Problem-Solving and Data Analysis | Ratios, rates, proportional relationships, and units | Easy <div><div></div><div></div><div></div></div> |

Question

In a box of pens, the ratio of black pens to red pens is **8** to **1**. There are **40** black pens in the box. How many red pens are in the box?

Answer

- A. **5**
- B. **8**
- C. **40**
- D. **320**

Correct Answer: A

Rationale

Choice A is correct. It’s given that the ratio of black pens to red pens is **8** to **1**. Therefore, there are  $\frac{1}{8}$  as many red pens as black pens in the box. It’s also given that there are **40** black pens in the box. Therefore, the number of red pens is  $\frac{1}{8}$  of the **40** black pens. Thus, the number of red pens is  $40\left(\frac{1}{8}\right)$ , or **5**.

Choice B is incorrect. This is the number of black pens in the box for every red pen.

Choice C is incorrect. This is the number of black pens in the box.

Choice D is incorrect. This is the number of red pens in the box if the ratio of black pens to red pens is **1** to **8**.